SQL Questions for assignment:

1.Find the 2nd highest salaried person in the finance department and also find his name?

Query:

```
SELECT first_name, last_name
FROM employees
WHERE emp_no = (
  SELECT emp_no
  FROM salaries
  WHERE salary = (
    SELECT MAX(salary)
    FROM salaries
    WHERE salary < (
      SELECT MAX(salary)
      FROM salaries
      WHERE emp_no IN (
        SELECT emp_no
        FROM dept_emp
        WHERE dept_no = (
          SELECT dept_no
          FROM departments
          WHERE dept_name = 'Finance'
        )
      )
    )
  )
);
```

Output:

Result Grid | Filter Rows:

first_name	last_name
Stella	Perry

2.Fetch the list of of employees who worked in Production department between 1998 and 2000?

Query:

```
SELECT emp_no, first_name, last_name
FROM employees
WHERE emp_no IN (
   SELECT emp_no
   FROM dept_emp
   WHERE dept_no = (
      SELECT dept_no
      FROM departments
      WHERE dept_name = 'Production'
   )
   AND from_date <= '2000-12-31'
   AND to_date >= '1998-01-01'
);
```

Output:

10003	Parto	Bamford
10004	Chirstian	Koblick
10010	Duangkaew	Piveteau
10018	Kazuhide	Peha
10020	Mayuko	Warwick
10024	Suzette	Pettey
10026	Yongqiao	Berztiss
10029	Otmar	Herbst
10030	Elvis	Demeyer
10032	Jeong	Reistad
10035	Alain	Chappelet
10044	Mingsen	Casley
10045	Moss	Shanbho...
10047	Zvonko	Nyanchama
10051	Hidefumi	Caine
10063	Gino	Leonhardt
10069	Margareta	Bierman
10081	Zhongwei	Rosen
10083	Vishv	Zockler
10084	Tuval	Kalloufi
10085	Kenroku	Malabarba

Note: The output is very long, so I have shown only a few sample records here. The remaining output is also available.

3.Who is the 3rd highest salaried person in Production department without using orderby.

Query:

```
SELECT emp_no, first_name, last_name
FROM employees
WHERE emp_no = (
  SELECT emp_no
  FROM salaries
  WHERE salary = (
    SELECT MAX(salary)
    FROM salaries
    WHERE salary < (
      SELECT MAX(salary)
      FROM salaries
      WHERE salary < (
        SELECT MAX(salary)
        FROM salaries
        WHERE emp_no IN (
          SELECT emp_no
          FROM dept_emp
          WHERE dept_no = (
            SELECT dept_no
            FROM departments
            WHERE dept_name = 'Production'
          )
        )
      )
    )
  )
);
```

Output:

Result Grid | Filter Rows:

emp_no	first_name	last_name
25472	Shrikanth	Hettesheimer
NULL	NULL	NULL

4.Find how many employees joined in which year in a complete single report.

Query:

```
SELECT YEAR(hire_date) AS join_year,
    COUNT(emp_no) AS total_employees
FROM employees
GROUP BY YEAR(hire_date);
```

Output:

Result Grid | Filter Rows:

join_year	total_employees
1986	36150
1985	35316
1989	28394
1994	14835
1990	25610
1992	20402
1987	33501
1995	12115
1993	17772
1999	1514
1991	22568
1988	31436
1997	6669
1996	9574
1998	4155
2000	13

5.Fetch the maximum employees joined year from the above report without using having.

Query:

```
SELECT join_year
FROM (
   SELECT YEAR(hire_date) AS join_year,
         COUNT(emp_no) AS total_employees
   FROM employees
   GROUP BY YEAR(hire_date)
) t
WHERE total_employees = (
   SELECT MAX(cnt) FROM ( SELECT COUNT(emp_no) AS cnt
      FROM employees
      GROUP BY YEAR(hire_date)
   ) x
);
```

Note: t is a **temporary name** given to the **result of the subquery**.

x is also a **temporary name**, for another subquery result.

Output:

Result Grid

	join_year
▶	1986

6.Need the reports total cost divided yearly in a table of all departments.

Query:

```
SELECT
   (SELECT dept_name
    FROM departments d
    WHERE d.dept_no = de.dept_no) AS department,
   YEAR(s.from_date) AS year,
   SUM(s.salary) AS total_cost
FROM salaries s, dept_emp de
WHERE s.emp_no = de.emp_no
GROUP BY de.dept_no, YEAR(s.from_date);
```

Output:

Result Grid | Filter Rows:

department	year	total_cost
Marketing	1993	262580338
Marketing	1994	291398019
Marketing	1995	320177206
Marketing	1996	351779856
Marketing	1997	382439137
Marketing	1998	410447130
Marketing	1999	440060991
Marketing	2000	443016308
Marketing	2001	441676726
Marketing	2002	257434866
Marketing	1992	233909442
Marketing	1988	111591698
Marketing	1989	141791060
Marketing	1990	170352794
Marketing	1991	200852152
Marketing	1987	81264653
Marketing	1986	55994427
Marketing	1985	27880162
Finance	1993	224286937
Finance	1994	249541547
Finance	1995	274617880

Note: The output is very long, so I have shown only a few sample records here. The remaining output is also available.

7.Find the Year where highest female employees joined.

Query:

```
SELECT join_year
FROM (
   SELECT YEAR(hire_date) AS join_year,
         COUNT(emp_no) AS female_count
   FROM employees
   WHERE gender = 'F'
   GROUP BY YEAR(hire_date)
) t
```

```
WHERE female_count = (
   SELECT MAX(cnt)
   FROM (
      SELECT COUNT(emp_no) AS cnt
      FROM employees
      WHERE gender = 'F'
      GROUP BY YEAR(hire_date)
   ) x
);
```

Output:

	join_year
▶	1986

8.Hire Date of the current manager of Human Resources department.(hint: to_Date year is 9999)

Query:

```
SELECT hire_date
FROM employees
WHERE emp_no = (
   SELECT emp_no
   FROM dept_manager
   WHERE dept_no = (
      SELECT dept_no
      FROM departments
      WHERE dept_name = 'Human Resources'
   )
   AND to_date = '9999-01-01'
);
```

Output:

	hire_date
▶	1985-08-04

9.Salary of the current manager of the Finance dept.

Query:

```
SELECT salary
FROM salaries
WHERE emp_no = (
   SELECT emp_no
   FROM dept_manager
   WHERE dept_no = (
      SELECT dept_no
      FROM departments
      WHERE dept_name = 'Finance'
   )
   AND to_date = '9999-01-01'
)
AND to_date = '9999-01-01';
```

Output:

Result Grid

	salary
▶	83457

10.Current dept managers first names along with emp_no.

Query:

```
SELECT emp_no, first_name
FROM employees
WHERE emp_no IN (
   SELECT emp_no
   FROM dept_manager
   WHERE to_date = '9999-01-01'
);
```

Output:

emp_no	first_name
110039	Vishwani
110114	Isamu
110228	Karsten
110420	Oscar
110567	Leon
110854	Dung
111133	Hauke
111534	Hilary
111939	Yuchang
NULL	NULL